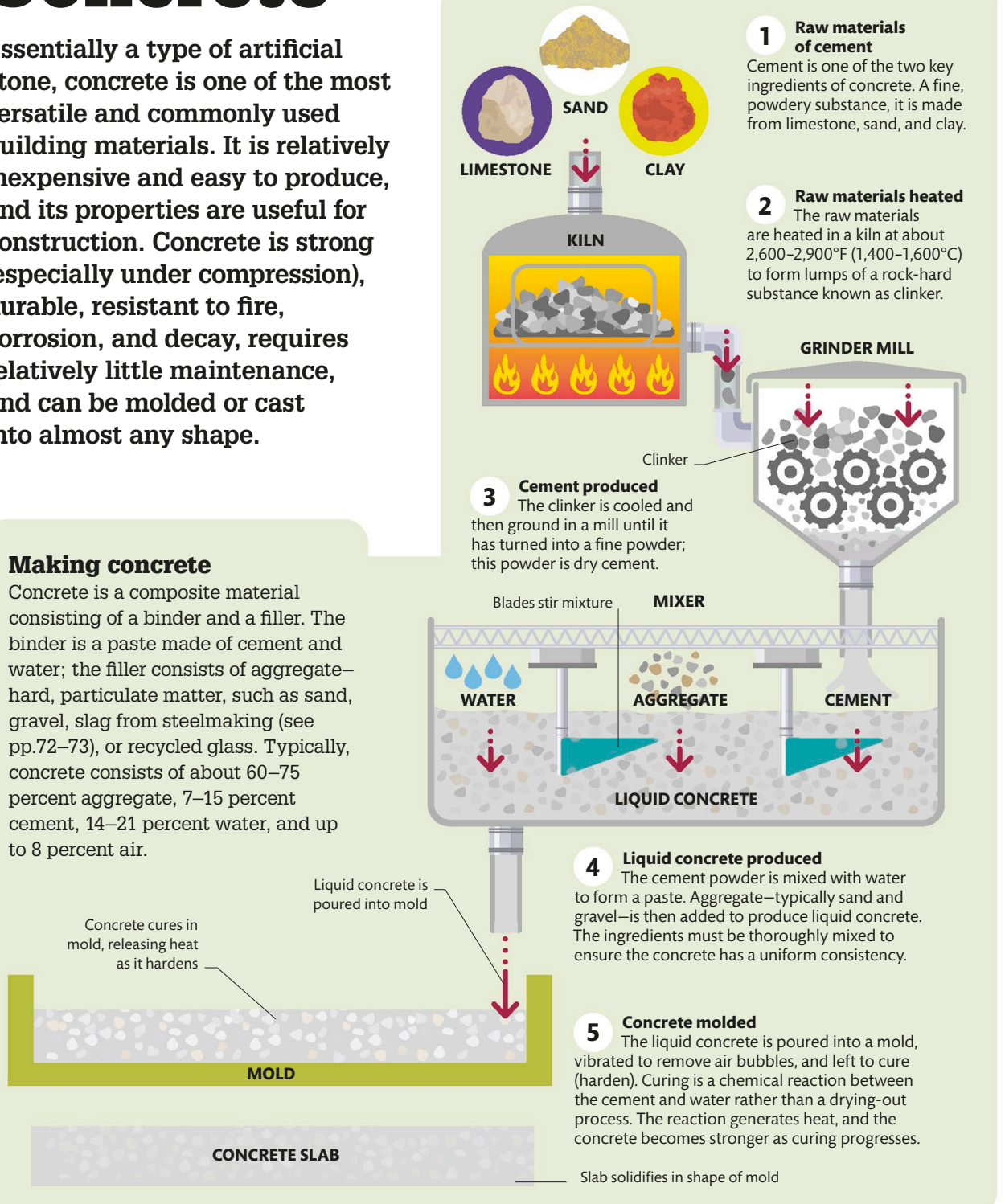
Concrete

Essentially a type of artificial stone, concrete is one of the most versatile and commonly used building materials. It is relatively inexpensive and easy to produce, and its properties are useful for construction. Concrete is strong (especially under compression), durable, resistant to fire, corrosion, and decay, requires relatively little maintenance, and can be molded or cast into almost any shape.

Making concrete

Concrete is a composite material consisting of a binder and a filler. The binder is a paste made of cement and water; the filler consists of aggregate—hard, particulate matter, such as sand, gravel, slag from steelmaking (see pp.72–73), or recycled glass. Typically, concrete consists of about 60–75 percent aggregate, 7–15 percent cement, 14–21 percent water, and up to 8 percent air.



1 Raw materials of cement
Cement is one of the two key ingredients of concrete. A fine, powdery substance, it is made from limestone, sand, and clay.

2 Raw materials heated
The raw materials are heated in a kiln at about 2,600–2,900°F (1,400–1,600°C) to form lumps of a rock-hard substance known as clinker.

3 Cement produced
The clinker is cooled and then ground in a mill until it has turned into a fine powder; this powder is dry cement.

4 Liquid concrete produced
The cement powder is mixed with water to form a paste. Aggregate—typically sand and gravel—is then added to produce liquid concrete. The ingredients must be thoroughly mixed to ensure the concrete has a uniform consistency.

5 Concrete molded
The liquid concrete is poured into a mold, vibrated to remove air bubbles, and left to cure (harden). Curing is a chemical reaction between the cement and water rather than a drying-out process. The reaction generates heat, and the concrete becomes stronger as curing progresses.